

Project

Introduction

This is an open-ended project. First, you will propose a brief project idea. After I approve your proposal, you will submit a short report describing what you accomplished and what you learned, along with any code you have written. Finally, you will present your work in class.

Generally, your project can fall into one of two categories:

System / Application Development:

Build a system that showcases a networking concept we've studied but haven't covered in a project yet. This could involve creating a networked application or using Software-Defined Networking (SDN) to manage network behavior. Your deliverables should include the application code and a report explaining what you built, how it works, and the concepts it demonstrates.

Network Analysis / Measurement:

Analyze network traffic in a specific scenario to understand network behavior. This may involve writing scripts to send packets, measuring network performance, or analyzing packet captures. Deliverables can include measurement results, such as figures or tables.

Please refer to the **Sample Topics** section for a list of potential ideas. Your idea does not need to strictly fit into one of the two suggested categories, as long as it is approved.

Groups

Groups should have two students. Exceptions can be made if:

1. If you have particular concerns, you may request to work on this project on your own. You should email me with your reasons.
2. A group of three students may be allowed with reasonable justification that the project requires it, along with a plan for distributing the work.

Implementation Requirements

You may use any programming language. Your implementation should show progress toward your goal and understanding of networking concepts. Use existing resources if needed, but document and understand them.

Project proposal (Wednesday, October 29 by 11:59pm ET)

Submit the [project proposal form](#) by the deadline. Briefly describe your project idea, expected deliverables, initial plan, and any questions on the Google form. You cannot propose the same project idea as someone else. Please check the Google Sheet first to avoid duplicates.

Final writeup and submission (Wednesday, December 3 by 11:59pm ET)

Submit your project by **pushing your code** to the repository and providing a **short write-up** (about 6-8 pages). The write-up should explain what you built and what you learned.

Your write-up should include:

- **Introduction:** State your project goals and summarize what you achieved.
- **Design/Implementation:** Describe what you built, how it works, and key design decisions. Include system components or measurement methods.
- **Discussion/Results:** Share your progress, results (e.g., logs, screenshots, figures), challenges, and lessons learned.
- **Conclusions/Future Work:** Summarize your learning, overall thoughts, potential next steps, and feedback on the project.

Submit what you have with a brief explanation of what you did and learned. Explore your interests and enjoy the project!

Presentation (December 8 and 10)

You will give an **in-class presentation** of your work.

Evaluation Criteria

- Technical Complexity and Completeness (30%)
- Problem-Solving and Creativity (30%)
- Experimental Analysis and Report Quality (30%)
- Presentation and Demonstration (10% of your grade)

Sample project topics

The following pages list sample project ideas. You can use, modify, or ignore them to develop your own ideas.

- Investigating Applications Traffic
 - WhatsApp vs iMessage vs Facebook Messenger
 - Zoom vs Google Meet vs Microsoft Teams

- Online Games
 - Live Streaming Traffic
 - Smart Home IoT Devices
- Network Simulation and Performance Analysis (Wireshark, Mininet, NS3)
 - Software-Defined Networking (SDN) vs Traditional
 - 5G vs 6G
 - DASH vs HLS
 - HTTP2 vs HTTP3
- Software-Defined Networking (SDN) Applications
 - Load Balancing
 - SDN-based DDoS Detection System
 - SDN-based Network Intrusion Detection System